AgriScore Project Work Plan

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Alternative Credit Scoring for Smallholder Farmers

Project Duration: May 8, 2026 – May 19, 2026

Prepared by: GROUP 3

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1. Project Goal

The goal of this project is to develop an MVP (Minimum Viable Product) for **AgriScore** — an alternative credit scoring system that predicts the likelihood of loan repayment among smallholder farmers using machine learning and alternative data sources such as mobile money transactions, agricultural productivity, and socioeconomic indicators.

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2. Project Objectives

By the end of the project, the team should be able to:

- Collect and prepare agricultural and financial datasets.
- Engineer useful features for credit scoring.
- Build and evaluate machine learning models.
- Create an explainable credit scoring system.
- Develop a dashboard for predictions and visualization.
- Present findings and recommendations.

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3. Project Timeline and Activities

Date	Activity	Description	Team Output
Friday, May 8	Project Kickoff & Planning	Team meeting to review proposal, define objectives, assign responsibilities.	Project roadmap, assigned roles
Saturday, May 9	Data Collection	Download LSMS-ISA datasets, Kaggle synthetic transaction datasets, and SAGDA agricultural datasets.	Raw datasets collected
Sunday, May 10			
Monday, May 11	Data Understanding & Exploration Data Cleaning & Preprocessing	Perform initial EDA Handle missing values, remove duplicates/outliers, encode categorical variables, scale numerical variables, and merge datasets.	EDA report and summary insights Cleaned and merged dataset
Tuesday, May 12	Feature Engineering	Create important features such as RFM metrics, yield stability, transaction frequency, cashflow volatility, and seasonal indicators.	Final feature dataset
Wednesday, May 13	Baseline Modeling	Train Logistic Regression model as the baseline model and evaluate initial performance.	Baseline model results
Thursday, May 14	Advanced Modeling	Train XGBoost and/or LightGBM models. Perform hyperparameter tuning and compare with baseline.	Improved model performance
Friday, May 15	Model Evaluation & Fairness Testing	Evaluate models using AUC-ROC, confusion matrix, precision-recall, Gini score, and fairness metrics. Generate SHAP explanations.	Final selected model
Saturday, May 16	Dashboard Development	Develop a simple Streamlit dashboard for entering farmer data and generating risk scores with explanations.	Working dashboard prototype

Date	Activity	Description	Team Output
Sunday, May 17	API & Integration (Optional Stretch Goal)	Develop a FastAPI endpoint for scoring predictions and test integration with dashboard.	API prototype
Monday, May 18	Documentation & Presentation Preparation	Prepare project documentation, methodology, findings, visualizations, and PowerPoint slides.	Final report and presentation slides
Tuesday, May 19	Final Review & Submission	Team review, debugging, testing, rehearsal of presentation, and final submission.	Completed project submission

4. Team Roles and Responsibilities

Role	Responsibilities
Project Manager	Coordinate tasks, track progress, ensure deadlines are met
Data Engineer	Collect, clean, and merge datasets
Data Analyst	Perform EDA and visualization
Machine Learning Engineer	Build and evaluate models
Backend/Deployment Lead	Develop Streamlit dashboard and API
Documentation & Presentation Lead	Prepare reports, slides, and final presentation

5. Tools and Technologies

Area	Tools
Programming	Python
Data Processing	pandas, numpy, polars
Visualization	matplotlib, seaborn, plotly

Area	Tools
Machine Learning	scikit-learn, XGBoost, LightGBM
Explainability	SHAP
Dashboard	Streamlit
API	FastAPI
Version Control	Git & GitHub
Collaboration	WhatsApp / Google Meet

6. Deliverables

- Cleaned and merged datasets
- Feature-engineered dataset
- Trained machine learning models
- Model evaluation report
- Explainability analysis using SHAP
- Streamlit dashboard
- Final presentation slides
- Final project report/documentation

7. Risk Management Plan

Risk	Possible Solution
Difficulty accessing real farmer data	Use synthetic/public datasets
Missing or inconsistent data	Apply imputation and preprocessing techniques
Model overfitting	Use cross-validation and tuning
Time constraints	Prioritize MVP features first
Team coordination issues	Hold short daily progress meetings

8. Success Criteria

The project will be considered successful if:

- The model achieves acceptable predictive performance (AUC > 0.78 target).
- The dashboard successfully generates farmer risk scores.
- The system provides explainable predictions.
- The project is completed and submitted on time.

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9. Conclusion

This work plan provides a structured roadmap for the successful development of the **AgriScore** project. By following this schedule, the team will be able to systematically collect data, build predictive models, evaluate performance, and deploy an MVP solution that supports financial inclusion for smallholder farmers in East Africa.

Task Tracker

Task	Assignee	Due Date	Status	Type
Project Kickoff & Planning		May 8, 2026	Done	Review
Data Collection		May 9, 2026	Done	Data
Data Understanding & Exploration (EDA)		May 10, 2026	Done	Data
Data Cleaning & Preprocessing		May 11, 2026	In Progress	Data
Feature Engineering		May 12, 2026	Not Started	Data
Baseline Modeling (Logistic Regression)		May 13, 2026	Not Started	Modeling
Advanced Modeling		May 14, 2026	Not Started	Modeling

Task	Assignee	Due Date	Status	Type
(XGBoost / LightGBM)				
Model Evaluation & Fairness Testing		May 15, 2026	Not Started	Modeling
Dashboard Development (Streamlit)		May 16, 2026	Not Started	Dashboard
API & Integration — FastAPI (Stretch Goal)		May 17, 2026	Not Started	Dashboard
Documentation & Presentation Preparation		May 18, 2026	Not Started	Documentation
Final Review & Submission		May 19, 2026	Not Started	Review
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